CHARACTERIZATION OF CATEGORIES FIBRED IN GROUPOIDS AMONG FUNCTORS

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Editorial note. Counted span is the author's own proof environment, source lines 7271-7315, following the statement at 7252-7269.

Editorial note. The final 'one checks' for chi = i^{-1} j is a direct verification from the displayed construction of i and j.

Allowed standard prerequisites (editorial).

- Definition of a category fibred in groupoids. Definition definition-fibred-groupoids of this chapter, conditions (1) and (2) used at proof lines 7289-7292 and 7297-7309. The two conditions whose interplay is the content of the equivalence; condition (2) is exactly strong cartesianness of every morphism.
- Pullback functors and the pseudo functor of a fibred category. Lemma lemma-fibred of this chapter, invoked at proof lines 7265-7267 and 7314. Supplies the data (U, f*, alpha) whose pseudo functoriality is the final statement.

Source reference. Original definition 003T: sources/stacks-categories/pinned-original-source.tex, source lines 7195–7195. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02XO: sources/stacks-categories/pinned-original-source.tex, source lines 6637–6637. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $p : \mathcal{S} \to \mathcal{C}$ be a functor. The following are equivalent*

(1) *$p : \mathcal{S} \to \mathcal{C}$ is a category fibred in groupoids, and*
(2) *all fibre categories are groupoids and $\mathcal{S}$ is a fibred category over $\mathcal{C}$.*

Moreover, in this case every morphism of $\mathcal{S}$ is strongly cartesian. In addition, given $f^*x \to x$ lying over f for all $f : V \to U = p(x)$ the data $(U \mapsto \mathcal{S}_U, f \mapsto f^*, \alpha_{f,g}, \alpha_U)$ constructed in Lemma 02XO defines a pseudo functor from $\mathcal{C}^{opp}$ in to the $(2,1)$-category of groupoids.

Proof. Assume $p : \mathcal{S} \to \mathcal{C}$ is fibred in groupoids. To show all fibre categories $\mathcal{S}_U$ for $U \in \mathrm{Ob}(\mathcal{C})$ are groupoids, we must exhibit for every $f : y \to x$ in $\mathcal{S}_U$ an inverse

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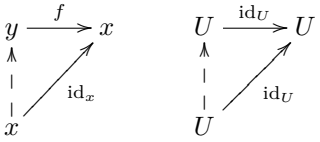
morphism. The diagram on the left (in $\mathcal{S}_U$) is mapped by p to the diagram on the right:



Since only id_U makes the diagram on the right commute, there is a unique $g : x \to y$ making the diagram on the left commute, so $fg = \mathrm{id}_x$. By a similar argument there is a unique $h : y \to x$ so that $gh = \mathrm{id}_y$. Then $fgh = f : y \to x$. We have $fg = \mathrm{id}_x$, so $h = f$. Condition (2) of Definition 003T says exactly that every morphism of $\mathcal{S}$ is strongly cartesian. Hence condition (1) of Definition 003T implies that $\mathcal{S}$ is a fibred category over $\mathcal{C}$.

Conversely, assume all fibre categories are groupoids and $\mathcal{S}$ is a fibred category over $\mathcal{C}$. We have to check conditions (1) and (2) of Definition 003T. The first condition follows trivially. Let $\phi : y \to x$, $\psi : z \to x$ and $f : p(z) \to p(y)$ such that $p(\phi) \circ f = p(\psi)$ be as in condition (2) of Definition 003T. Write $U = p(x)$, $V = p(y)$, $W = p(z)$, $p(\phi) = g : V \to U$, $p(\psi) = h : W \to U$. Choose a strongly cartesian $g^*x \to x$ lying over g. Then we get a morphism $i : y \to g^*x$ in $\mathcal{S}_V$, which is therefore an isomorphism. We also get a morphism $j : z \to g^*x$ corresponding to the pair (ψ, f) as $g^*x \to x$ is strongly cartesian. Then one checks that $\chi = i^{-1} \circ j$ is a solution.

We have seen in the proof of (1) $\Rightarrow$ (2) that every morphism of $\mathcal{S}$ is strongly cartesian. The final statement follows directly from Lemma 02XO. $\square$